AI-Lingo

Contextual Language Immersion App

USER REQUIREMENTS DOCUMENT

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Document scope	Stakeholder Analysis, User Stories, and Use Cases
Project focus	Personalized AI-generated language conversations

This document defines the key stakeholders, user-facing needs, and main interaction flows for AI-Lingo. It is intended to support the next stage of system requirements analysis.

1. Stakeholder Analysis

Stakeholder analysis identifies the people and external systems that interact with the application or influence its functionality.

Stakeholder	Role	Needs and Expectations
Language Learner	Primary user	Personalized conversations, suitable difficulty, vocabulary practice, and progress tracking.
Teacher / Language Instructor	Learning support	Learners practice realistic conversations and improve vocabulary and language skills.
Application Administrator	Application maintenance	Manage system settings, monitor performance, and handle technical issues.
AI Service Provider	External supporting system	Receive appropriate prompts and generate relevant dialogues that match the learner's level.

Primary stakeholder: The language learner, because the application is designed around their learning goals, interests, and proficiency level.

2. User Stories

User stories describe required functionality from the learner's perspective.

ID	User Story	Priority
US-01	As a learner, I want to Select a target language to practice the language I want to learn.	High
US-02	As a learner, I want to Specify my proficiency level so conversations match my abilities.	High
US-03	As a learner, I want to Choose topics such as business meetings or ordering at a café so practice is relevant to my life.	High
US-04	As a learner, I want to Generate contextual conversations so I can practice realistic scenarios.	High
US-05	As a learner, I want to Constrain AI vocabulary to words I have already learned, so I am not overwhelmed.	High
US-06	As a learner, I want to Interact with generated dialogues to practice using vocabulary in context.	High
US-07	As a learner, I want to Save new vocabulary so I can review it later.	High
US-08	As a learner, I want to Track my learning progress so I can understand my improvement over time.	High
US-09	As a learner, I want to Access saved lesson snippets offline so I can review without internet.	Medium
US-10	As a learner, I want to Keep my personal learning data protected and private.	High
US-11	As a learner, I want to Resume an active conversation without losing available progress.	Medium
US-12	As a learner, I want to Review previously learned vocabulary to reinforce retention.	Medium

3. Use Cases

Use cases describe the interaction between the learner and the application to achieve a specific goal.

UC-01: Set Up Learning Preferences

Primary actor	Language Learner
Goal	Configure language-learning preferences.
Preconditions	The application is installed and accessible.
Main flow	1. The learner opens the application. 2. The learner selects a target language. 3. The learner specifies their proficiency level. 4. The learner selects interests or preferred conversation topics. 5. The application saves the preferences.
Alternative flows	• If optional preferences are skipped, the application may use default topics or request missing information when needed.
Postconditions	The learner's preferences are available for personalized lesson generation.

UC-02: Generate a Personalized Conversation

Primary actor	Language Learner; AI Service supports the flow
Goal	Generate a realistic conversation suited to the learner's level and interests.
Preconditions	A target language and conversation topic have been selected.
Main flow	1. The learner selects a scenario, such as ordering at a café. 2. The application retrieves the proficiency level and relevant learned vocabulary. 3. The application constructs a prompt with the scenario, level, and vocabulary constraints. 4. The AI service generates a contextual dialogue. 5. The application displays the dialogue. 6. The learner reviews the conversation and begins practicing.
Alternative flows	• If the AI service is unavailable, inform the learner and offer a saved lesson if one is available. • If the dialogue violates required constraints, retry generation or display an error.
Postconditions	A suitable conversation is available for practice.

UC-03: Practice a Conversation

Primary actor	Language Learner
Goal	Practice the target language through contextual dialogue.
Preconditions	A generated or saved conversation is available.
Main flow	1. The learner opens a conversation. 2. The application displays the dialogue. 3. The learner practices the presented expressions. 4. The learner identifies unfamiliar vocabulary. 5. The learner saves useful words for later review. 6. The application updates relevant learning data.
Alternative flows	• If the learner leaves the session, preserve available progress where supported.
Postconditions	The learner has practiced the dialogue, and saved vocabulary is available for review.

UC-04: Save and Review Vocabulary

Primary actor	Language Learner
Goal	Build and review a personal vocabulary collection.
Preconditions	A lesson or conversation containing vocabulary is available.
Main flow	1. The learner selects a word or expression. 2. The learner chooses to save it. 3. The application stores the vocabulary locally. 4. The learner opens the saved vocabulary section. 5. The application displays saved words. 6. The learner reviews words and available contextual examples.
Alternative flows	• If the word is already saved, avoid a duplicate or inform the learner.
Postconditions	Selected vocabulary is stored and can be reviewed later.

UC-05: Track Learning Progress

Primary actor	Language Learner
Goal	Monitor learning activity and progress.
Preconditions	At least one learning activity has been completed.
Main flow	1. The learner opens the progress section. 2. The application retrieves stored learning records. 3. The application displays available metrics, such as completed lessons, saved vocabulary, and practice activity. 4. The learner reviews their learning history.
Alternative flows	• If no activity has been recorded, display an empty state and encourage the learner to start a lesson.
Postconditions	The learner can view recorded learning activity.

UC-06: Access Lessons Offline

Primary actor	Language Learner
Goal	Review previously saved content without internet access.
Preconditions	Lesson snippets or vocabulary have been saved on the device.
Main flow	1. The learner opens the application without an internet connection. 2. The application retrieves locally stored lessons and vocabulary. 3. The learner opens a saved item. 4. The application displays the available content.
Alternative flows	• If the requested content is not stored locally, inform the learner that internet access may be required.
Postconditions	The learner can access the saved content offline.

4. User Requirements Summary

Personalization: Allow learners to select a target language, proficiency level, and preferred topics.

AI conversation generation: Generate realistic contextual dialogues using prompt chaining and suitable constraints.

Vocabulary management: Allow learners to save and review learned vocabulary.

Progress tracking: Record and display relevant learning activity and progress.

Offline access: Store selected vocabulary and lesson snippets locally using Hive NoSQL.

Data protection: Protect sensitive learning data using appropriate security mechanisms, including secure key management where applicable.

5. Assumptions and Scope

• The initial version focuses on text-based AI-generated conversations unless additional interaction modes are agreed upon by the team.

• Internet access is required for AI conversation generation unless an offline generation capability is implemented.

• Offline functionality is limited to content previously stored on the device.

• Supported languages, proficiency levels, and progress metrics will be finalized during system requirements analysis.

• Authentication, teacher dashboards, and administrative features should only be included if confirmed within the agreed project scope.

Next step: Use this document as the basis for defining the System Requirements, including functional and non-functional requirements.